- All problems are worth an equal weight, even if some may be harder than others.
- Write answers on a separate sheet, either typed or handwritten, either on paper or a note taking device. No need to copy the problem statements. Please, upload an unique pdf file and assign each question one (or multiple) pages of your upload file.
- Late submissions will be worth significantly less points.
- Be clear about your final answer, either circling it or stating it.
- It is okay to work with others. In fact, I encourage you to work together. However, acknowledge them in your paper and write your solutions in your own words. You are not learning anything by copying from another source.

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1. Consider the system given by

$$\begin{cases} x' = 25y - 13x \\ y' = 17y - 9x \end{cases}$$

a) Verify that both

$$X_1(t) = \begin{bmatrix} 5e^{2t} \\ 3e^{2t} \end{bmatrix} \quad \text{and} \quad X_2(t) = \begin{bmatrix} e^{2t}(5t+3) \\ e^{2t}(3t+2) \end{bmatrix}$$

are solutions.

b) Can you use $X_1(t)$ and $X_2(t)$ to find the solution for the initial conditions

$$x(0) = 1\ ,\ y(0) = 1?$$

If so, show how and cite the name of the principle being used.

c) Find the solution for the initial conditions

$$x(0) = y(0) = 0.$$

2. Consider the system given by

$$\begin{cases} x' = x - y \\ y' = x + 3y \end{cases}$$

a) Verify that

$$X_1(t) = \begin{bmatrix} te^{2t} \\ -(t+1)e^{2t} \end{bmatrix}$$

is a solution to the system.

b) Even without knowing another solution $X_2(t)$, it is possible to find the solution with initial values

$$x(0) = 0, \quad y(0) = 5.$$

What is the solution?

3. Consider the system

$$\begin{cases} x' = y + 2 \\ y' = 2x + y - 4 \end{cases}$$

that cannot be written in the form $X' = AX$.

a) Write the system in the form $X' = BX + b$, where B is a 2×2 matrix and b is a 2×1 constant vector.

b) Introduce the variables

$$u = x - 3$$
$$v = y + 2$$

and write a new system that has only the new variables

$$\begin{cases} u' = f(u, v) \\ v' = g(u, v) \end{cases}$$

c) Write the new system in matrix form $Y' = AY$.

d) Compute the resulting matrix $A - B$.

e) Suppose that a system of differential equations is written as

$$x' = ax + by + r$$
$$y' = cx + dy + s$$

and suppose that there exists an unique equilibrium point (p, q). Which new variables u, v must be introduced to make the new system

$$u' = f(u, v)$$
$$v' = g(u, v)$$

be writable as $X' = AX$? What is A?

4. Draw all straight-line solutions on the phase plane of the system given by $X' = AX$ for each matrix A given.

a) $A = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$ b) $A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$ c) $A = \begin{bmatrix} 4 & -4 \\ 1 & 0 \end{bmatrix}$

5. Solve all initial value problems

a) $\begin{cases} x' = -2x - 2y \\ y' = -2x + y \\ x(0) = 1 \\ y(0) = -2 \end{cases}$ b) $\begin{cases} x' = 3x \\ y' = x - 2y \\ x(0) = 2 \\ y(0) = 2 \end{cases}$ c) $\begin{cases} x' = 4x - 2y \\ y' = x + y \\ x(0) = -1 \\ y(0) = -2 \end{cases}$

6. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$$

a) Compute e^A, e^B and e^{A+B}.

b) Verify that $e^A e^B \neq e^{A+B}$.

7. Classify the kind of equilibrium at the origin for $X' = AX$ for each matrix below.

$$\text{a)} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \quad \text{b)} \begin{bmatrix} 10 & -5 \\ 13 & -6 \end{bmatrix} \quad \text{c)} \begin{bmatrix} 0 & -2 \\ 8 & 0 \end{bmatrix} \quad \text{d)} \begin{bmatrix} 17 & -30 \\ 9 & -16 \end{bmatrix}$$

$$\text{e)} \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix} \quad \text{f)} \begin{bmatrix} 42 & 40 \\ 0 & 2 \end{bmatrix} \quad \text{g)} \begin{bmatrix} -42 & 0 \\ 80 & -3 \end{bmatrix} \quad \text{h)} \begin{bmatrix} 42 & 84 \\ 21 & 42 \end{bmatrix}$$

$$\text{i)} \begin{bmatrix} 6 & -3 \\ 2 & -1 \end{bmatrix} \quad \text{j)} \begin{bmatrix} 5 & 4 \\ -10 & -7 \end{bmatrix} \quad \text{k)} \begin{bmatrix} 8 & -1 \\ 4 & 4 \end{bmatrix} \quad \text{l)} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

8. Let $a, b \in \mathbb{R}$ be two real parameters, and consider the system given by

$$\begin{cases} x' = ax - by \\ y' = -bx + ay \end{cases}$$

Draw the regions of the parameter plane (a, b)-plane in which the values of a and b give rise to

- a saddle point;
- a source;
- a sink.

9. Let $a \in \mathbb{R}$ be a real parameter, and consider the system given by

$$\begin{cases} x' = ax + (4 - 3a)y \\ y' = -ax + 3ay \end{cases}$$

Draw on the trace-determinant plane the curve described by the coefficients matrix as we increase the parameter a from $-\infty$ to ∞, and describe the type of system that we can have depending on the parameter a.

10. Convert the harmonic oscillator equation with the corresponding parameters into a 2×2 system and use eigenvalue and eigenvector analysis to obtain a sketch of the phase portrait. Compare your results with your MATLAB assignment.

a) $m = 1$, $b = 5$, $k = 4$

b) $m = 1$, $b = 4$, $k = 4$

c) $m = 1$, $b = 3$, $k = 4$